

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06 <i>04</i>	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	<i>G. Hariprasad</i>	Reg. No.:	<i>192525239</i>
Section / Batch:	<i>AI & ML (B)</i>	Date:	<i>24/07/2026</i>

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I Gu Han: prasad (Name / Reg No) 192525273 certify that this submission is my original work and that I have 2/19 adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Gu Han: prasad

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

convex Hull - Rectangular Boundary with interior points

=> Given points

$$S = \{ (0,0), (0,5), (5,5), (5,0), (2,2), (3,3), (1,4) \}$$

Problem Interpretation

The objective is to find the convex polygon that encloses all the given points. The points on the outer boundary form the convex hull, while the points inside are excluded.

=> Brute Force convex Hull Algorithm

- 1) considers every pair of points
- 2) draw a line joining the pair.
- 3) check the position of all remaining points
- 4) if all remaining points lie on the same side of the line, the line is convex Hull edge.

≤ otherwise, rejected the line.

practical significance.

- Easy to understand and implement.
- suitable for small numbers of points
- not efficient for large datasets because the number of comparisons increases rapidly

⇒ conclusion:

using the brute force Hull algorithm.

The outer boundary consists the corner points

$(0,0)$ $(0,5)$ $(5,5)$ and $(5,0)$ - the interior points

$(2,2)$ $(3,3)$ and $(4,4)$ are exclude. 105 comparisons,

time complexity of $O(n^3)$

7C

convex hull - Boundary detection in urban layout

Given point

$S = \{ (0,0), (2,5), (5,7), (8,4), (7,0), (4,2), (3,3) \}$

problem interpretation

The objective is to identify the convex hull, which is the smallest convex polygon enclosing all the given points in an urban layout. The convex hull represents the outer boundary of the marked area.

(2)

Rule. Force convex Hull Algorithm

- 1) consider every pair of points
- 2) draw a line joining the pair.
- 3) Check whether all remaining points lie on the same side of the line.
- if they do, the line is a convex hull edge otherwise reject it.

convex. Hull edges

→ The following edges form convex hull.

• $(0,0) - (2,5)$ ✓

• $(2,5) - (5,7)$ ✓

• $(5,7) - (8,4)$ ✓

• $(8,4) - (7,0)$ ✓

• $(7,0) - (0,0)$ ✓

These edges form the outer boundary point set

interior

points

the points

$(1,2)$

$(3,3)$

inside the convex boundary not include in hull

Final convex Hull

$\{(0,0), (2,5), (5,7), (8,4), (7,0)\}$

number of comparisons

- number of points, $n=7$
- possible line pairs $= 7 \times 6/2 = 21$
- remaining points checked for each pair $= 5$

Total comparisons $= 21 \times 5 = 105$

Time complexity of brute force convex Hull algorithm

checks every pair of points and compares them with all remaining points

Time complexity $= O(n^3)$

⇒ practical significance

- identifies the outer boundary of urban region
- useful in urban mapping, GIS city land boundary detection.
- suitable for small datasets but inefficient for large datasets because of its $O(n^3)$ complexity.

conclusion

⇒ using the Quick Sort Convex Hull algorithm.

boundary points $(0,0)$, $(2,5)$, $(5,7)$, $(8,4)$ and $(7,0)$ while $(4,2)$ and $(3,3)$ are

interior points. The algorithm performs

comparisons has time complexity $O(n^2)$